

```

# STUDENT PERFORMANCE ANALYZER

## PROJECT REPORT
### Project Title: Student Performance Analyzer
### Submitted By: Darakhsha Anjum
### Course: B.Tech Data Science
### Year: 2nd Year
### College Name: Goel Institute Of Technology and Management ## Introduction
### Student Performance Analyzer is a Python-based project that analyzes students'
academic performanc.
### This project uses the Pandas library to process student data and calculate
percentages, assign grades, identify toppers,
### and evaluate overall class performance.It also provides data visualization
through graphs for better understanding and
### analysis of student results.
## Objectives

# STUDENT MARKS ANALYZE

## ALGORITHM
### 1. Start the program.
### 2. Enter student name.
### 3. Enter marks of all subject.
### 4. Calculate total marks.
### 5. Calculate average marks.
### 6. Calculate percentage.
### 7. Find the highest and lowest marks.
### 8. Check pass/fail status.
### 9. Display the result.
### 10. End the program.

# Python cod
Name = input("Enter Student Name: ")
m1 = float(input("Enter marks of Student 1: ")) m2 =
float(input("Enter Marks of Student 2: ")) m3 =
float(input("Enter marks of Student 3: ")) total = m1 + m2 + m3
average = total / 3
percentage = (total / 300) * 100
highest = max(m1, m2, m3)
lowest = min(m1, m2, m3)
print("\n--- Student Marks Analysis ---")
print("Student Name:")
print("Total Marks:", total)
print("Average Marks:", average)
print("Percentage:", percentage, "%")
print("Highest Marks:", highest)
print("Lowest Marks:", lowest)
if m1 >= 33 and m2 >= 33 and m3 >= 33:
    print("Result: Pass")
else:

```

```

print("Result: Fail")

# output

# MARKS: 80,75,90
# TOTAL = 245
# AVERAGE = 81.67
# PERCENTAGE = 81.67%
# HIGHEST = 90
# LOWEST = 75
# RESULT = PASS

# GRAPHS KE THROUGH DATA VISUALIZE
import matplotlib.pyplot as plt
student = ["Ali", "sara", "ahmed", "Ayesha"]
marks = [80, 75, 90, 85]
plt.bar(student, marks)
plt.title("Student Marka Analysis")
plt.xlabel("Student")
plt.ylabel("Marks")
plt.savefig("student_chart.png")
plt.show()

# TOOLS USED

### 1. Python
### 2. Pandas
### 3. Matplotlib
### 4. PyCharm
### 5. CSV File

# DATASET
# Python Code

data = {
'Student': ['Ali', 'Sara', 'Ahmed', 'Ayesha'],
'Math': [85, 92, 70],
'Science': [78, 88, 75],
'English': [90, 95, 80]
}
df = pd.DataFrame(data)
print(df)

# Complete Python Code

import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("studen.csv")
df["Total"] = df["Maths"] + df["Science"] + df["English"] df["Percentage"] =
df["Total"] / 3
def grade(per):
    if per >= 90:
        return "A+"
    elif per >= 80:
        return "A"
    elif per >= 70:

```

```

        return "B"
    elif per >= 60:
        return "C"
    else:
        return "F"
    df["Grade"] = df["Percentage"].apply(grade)
    print(df)
    topper = df.loc[df["Percentage"].idxmax()]
    print("\nTopper:", topper["Name"])
    average = df["Percentage"].mean()
    print("Class Average:", average)
    plt.bar(df["Name"], df["Percentage"])
    plt.title("Student Percentage")
    plt.xlabel("Student")
    plt.ylabel("Percentage")
    plt.show()

# OUTPUT
## NAME      PERCENTAGE      GRADE
### Ali      81.67          A
### Sara     91.67          A+
### Ahmed    65.00          C
### Ayesha    85.00          A
### Topper: Sara
### Class Average: 80.83

# GRAPH
### The bar graph displays the percentage of each student.

# ADVANTAGES
###. Fast calculation
###. Accurate results
###. Easy data analysis
###. Visual representation through graphs

# CONCLUSION
###The Student Perfotmance Analyzer project uses Python,Pandas,and Matplotlib to
analyze student performance.
### This project helps in understanding basic concepts of data analysis and data
visulization.
### It provides quick and accurate results and presents student performance in a
graphical format for better
###understanding.

```